

22 Window watchdog (WWDG)

This section applies to the whole STM32F4xx family, unless otherwise specified.

22.1 WWDG introduction

The window watchdog is used to detect the occurrence of a software fault, usually generated by external interference or by unforeseen logical conditions, which causes the application program to abandon its normal sequence. The watchdog circuit generates an MCU reset on expiry of a programmed time period, unless the program refreshes the contents of the downcounter before the T6 bit becomes cleared. An MCU reset is also generated if the 7-bit downcounter value (in the control register) is refreshed before the downcounter has reached the window register value. This implies that the counter must be refreshed in a limited window.

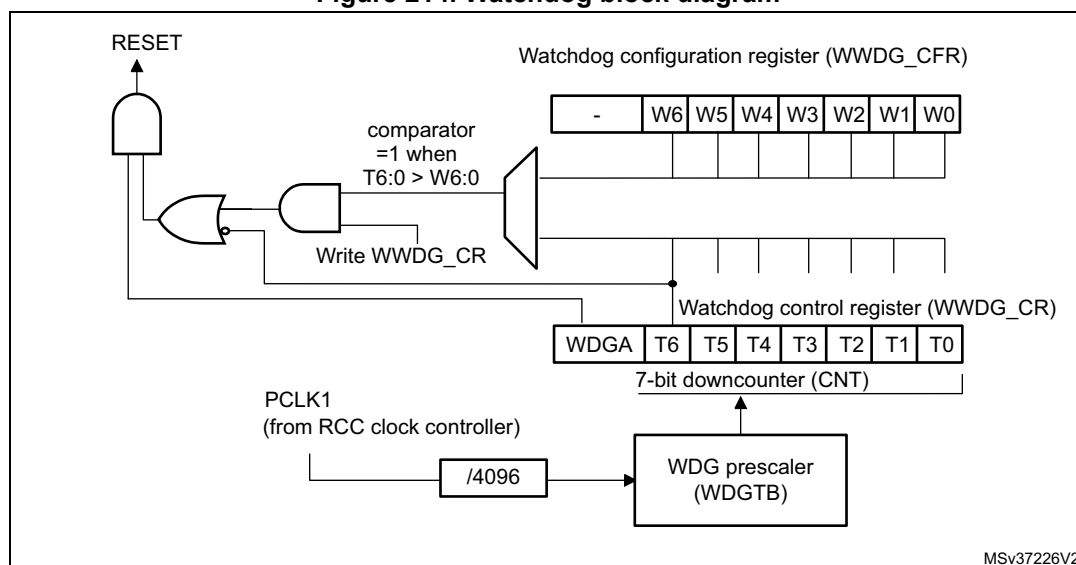
22.2 WWDG main features

- Programmable free-running downcounter
- Conditional reset
 - Reset (if watchdog activated) when the downcounter value becomes less than 0x40
 - Reset (if watchdog activated) if the downcounter is reloaded outside the window (see [Figure 215](#))
- Early wake-up interrupt (EWI): triggered (if enabled and the watchdog activated) when the downcounter is equal to 0x40.

22.3 WWDG functional description

If the watchdog is activated (the WDGA bit is set in the WWDG_CR register) and when the 7-bit downcounter (T[6:0] bits) rolls over from 0x40 to 0x3F (T6 becomes cleared), it initiates a reset. If the software reloads the counter while the counter is greater than the value stored in the window register, then a reset is generated.

Figure 214. Watchdog block diagram



The application program must write in the WWDG_CR register at regular intervals during normal operation to prevent an MCU reset. This operation must occur only when the counter value is lower than the window register value. The value to be stored in the WWDG_CR register must be between 0xFF and 0xC0.

Enabling the watchdog

The watchdog is always disabled after a reset. It is enabled by setting the WDGA bit in the WWDG_CR register, then it cannot be disabled again except by a reset.

Controlling the downcounter

This downcounter is free-running, counting down even if the watchdog is disabled. When the watchdog is enabled, the T6 bit must be set to prevent generating an immediate reset.

The T[5:0] bits contain the number of increments which represents the time delay before the watchdog produces a reset. The timing varies between a minimum and a maximum value due to the unknown status of the prescaler when writing to the WWDG_CR register (see [Figure 215](#)). The Configuration register (WWDG_CFR) contains the high limit of the window: To prevent a reset, the downcounter must be reloaded when its value is lower than the window register value and greater than 0x3F. [Figure 215](#) describes the window watchdog process.

Note: The T6 bit can be used to generate a software reset (the WDGA bit is set and the T6 bit is cleared).

Advanced watchdog interrupt feature

The Early Wake-up Interrupt (EWI) can be used if specific safety operations or data logging must be performed before the actual reset is generated. The EWI interrupt is enabled by setting the EWI bit in the WWDG_CFR register. When the downcounter reaches the value 0x40, an EWI interrupt is generated and the corresponding interrupt service routine (ISR) can be used to trigger specific actions (such as communications or data logging), before resetting the device.

In some applications, the EWI interrupt can be used to manage a software system check and/or system recovery/graceful degradation, without generating a WWDG reset. In this case, the corresponding interrupt service routine (ISR) should reload the WWDG counter to avoid the WWDG reset, then trigger the required actions.

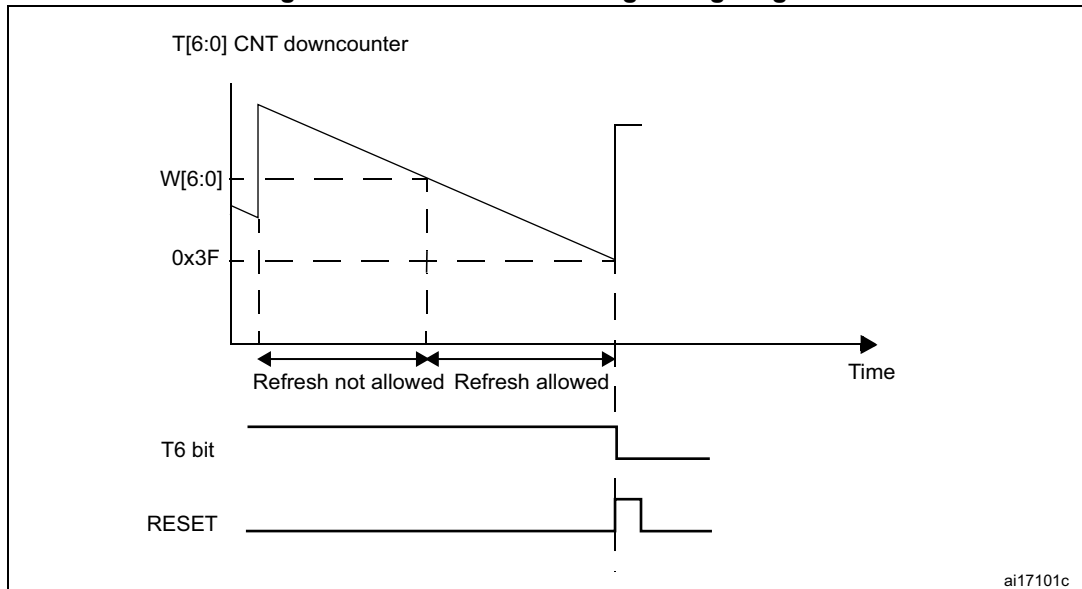
The EWI interrupt is cleared by writing '0' to the EWIF bit in the WWDG_SR register.

Note: *When the EWI interrupt cannot be served (due to a system lock in a higher priority task), the WWDG reset is eventually generated.*

22.4 How to program the watchdog timeout

Warning: When writing to the WWDG_CR register, always write 1 in the T6 bit to avoid generating an immediate reset.

Figure 215. Window watchdog timing diagram



The formula to calculate the WWDG timeout value is given by:

$$t_{\text{WWDG}} = t_{\text{PCLK1}} \times 4096 \times 2^{\text{WDGTB}[1:0]} \times (T[5:0] + 1) \quad (\text{ms})$$

where:

t_{WWDG} : WWDG timeout

t_{PCLK1} : APB1 clock period measured in ms

4096: value corresponding to internal divider

As an example, let us assume APB1 frequency is equal to 24 MHz, WDG TB[1:0] is set to 3 and T[5:0] is set to 63:

$$t_{\text{WWDG}} = 1 / 24000 \times 4096 \times 2^3 \times (63 + 1) = 21.85\text{ms}$$

Refer to [Table 109](#) for the minimum and maximum values of the t_{WWDG} .

Table 109. Minimum and maximum timeout values at 30 MHz (f_{PCLK1})

Prescaler	WDGTB	Min timeout (μs) T[5:0] = 0x00	Max timeout (ms) T[5:0] = 0x3F
1	0	136.53	8.74
2	1	273.07	17.48
4	2	546.13	34.95
8	3	1092.27	69.91

22.5 Debug mode

When the microcontroller enters debug mode (Cortex[®]-M4 with FPU core halted), the WWDG counter either continues to work normally or stops, depending on DBG_WWDG_STOP configuration bit in DBG module. For more details, refer to [Section 38.16.2: Debug support for timers, watchdog, bxCAN and I²C](#).

22.6 WWDG registers

Refer to [Section 2.2 on page 45](#) for a list of abbreviations used in register descriptions.

The peripheral registers have to be accessed by half-words (16 bits) or words (32 bits).

22.6.1 Control register (WWDG_CR)

Address offset: 0x00

Reset value: 0x0000 007F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								WDGA	T[6:0]						
								rs	rw						

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **WDGA**: Activation bit

This bit is set by software and only cleared by hardware after a reset. When WDGA = 1, the watchdog can generate a reset.

0: Watchdog disabled

1: Watchdog enabled

Bits 6:0 **T[6:0]**: 7-bit counter (MSB to LSB)

These bits contain the value of the watchdog counter. It is decremented every $(4096 \times 2^{\text{WDGTB}[1:0]})$ PCLK1 cycles. A reset is produced when it rolls over from 0x40 to 0x3F (T6 becomes cleared).

22.6.2 Configuration register (WWDG_CFR)

Address offset: 0x04

Reset value: 0x0000 007F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						EWI		WDGTB[1:0]		W[6:0]					
						rs		rw		rw					

Bit 31:10 Reserved, must be kept at reset value.

Bit 9 **EWI**: Early wake-up interrupt

When set, an interrupt occurs whenever the counter reaches the value 0x40. This interrupt is only cleared by hardware after a reset.

Bits 8:7 **WDGTB[1:0]**: Timer base

The time base of the prescaler can be modified as follows:

00: CK Counter Clock (PCLK1 div 4096) div 1

01: CK Counter Clock (PCLK1 div 4096) div 2

10: CK Counter Clock (PCLK1 div 4096) div 4

11: CK Counter Clock (PCLK1 div 4096) div 8

Bits 6:0 **W[6:0]**: 7-bit window value

These bits contain the window value to be compared to the downcounter.

22.6.3 Status register (WWDG_SR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														EWIF	
														rc_w0	

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **EWIF**: Early wake-up interrupt flag

This bit is set by hardware when the counter has reached the value 0x40. It must be cleared by software by writing '0'. A write of '1' has no effect. This bit is also set if the interrupt is not enabled.

22.6.4 WWDG register map

The following table gives the WWDG register map and reset values.

Table 110. WWDG register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0x00	WWDG_CR	Reserved																							WDGA	T[6:0]								
	Reset value																								0	1	1	1	1	1	1	1		
0x04	WWDG_CFR	Reserved																							EWI	WDGTB1	WDGTB0	W[6:0]						
	Reset value																								0	0	0	1	1	1	1	1	1	1
0x08	WWDG_SR	Reserved																																EWIF
	Reset value																																	0

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.